
PRACTICE EXAM 3 (FORM A)

NAME: _____

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Answer the following questions concerning spanning sets.

- (a) Let V be a vector space and suppose $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a set of vectors in V . State precisely what it means for the set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ to be a spanning set for V .

- (b) Determine if the set $\{(2, 1, 0), (0, 2, 3), (-3, 5, 0)\}$ is a spanning set for $\mathbb{R}^3$.

Problem 2. Consider the matrix $A = \begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$.

(a) Compute the characteristic polynomial $c_A(x)$ and find the only eigenvalue λ of A .

(b) For the eigenvalue λ that you obtained above, find a basis of the eigenspace of A associated with λ . Use this to decide if A is diagonalizable or not.

Problem 3. Let $\mathbf{P}_2$ be the space of polynomials of degree at most 2 (including the zero polynomial). Define a transformation $T: \mathbf{P}_2 \rightarrow \mathbf{P}_2$ via $T(p(x)) = p'(x) - 2p(0)$.

(a) Show that T is a linear transformation.

(b) Compute bases for $\ker T$ and $\operatorname{im} T$.

Problem 4. Let V be a vector space. Show that the following statements are true using the axioms of vector spaces:

(a) The zero vector is unique. That is, if $\mathbf{0}, \mathbf{0}' \in V$ are two zero vectors in V , then $\mathbf{0} = \mathbf{0}'$.

(b) If $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ are vectors such that $\mathbf{u} + \mathbf{w} = \mathbf{v} + \mathbf{w}$, then $\mathbf{u} = \mathbf{v}$.

(c) If $\mathbf{x} \in V$ is a vector, then $0\mathbf{x} = \mathbf{0}$.

Scratch working page

Scratch working page

PRACTICE EXAM 3 (FORM B)

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INSTRUCTIONS

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- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Answer the following questions concerning similar matrices.

- (a) Let A and B be two square matrices of the same size. State precisely what it means for A to be similar to B . Moreover, prove that if A is similar to B , then $\det(A) = \det(B)$.

- (b) Show that the matrices

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 5 & 0 \\ 0 & -1 & 0 \\ 2 & -1 & -2 \end{bmatrix}$$

are not similar.

Problem 2. Consider the matrix

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and define the set

$$U = \{A \in \mathbf{M}_{22} \mid AJ = JA\} \subseteq \mathbf{M}_{22}.$$

(a) Prove that U is a vector subspace of $\mathbf{M}_{22}$.

(b) Compute a basis and the dimension of U .

Problem 3. Let

$$A = \begin{bmatrix} 0 & 2 & -1 & 1 \\ 3 & 1 & 10 & 5 \\ 1 & 0 & 3 & 1 \end{bmatrix}.$$

(a) Find a basis and the dimension of the subspaces $\text{row } A$ and $\text{col } A$.

(b) Compute $\dim(\text{null } A)$ without finding a basis of $\text{null } A$.

Problem 4. Decide whether each of the statements below is true or false. Completely fill the circle that is to the left of your choice. You do not need to justify your choices.

- (a) If $T: V \rightarrow W$ is a linear transformation between two finite dimensional vector spaces, then $\dim(\ker T) + \dim(\operatorname{im} T) = \dim W$.

☐ True

☐ False

- (b) If V is a vector space and $S \subseteq V$ is a spanning set for V , then S contains a basis of V .

☐ True

☐ False

- (c) The vector space $\mathbb{R}^5$ admits a spanning set with three elements.

☐ True

☐ False

- (d) If A is an $n \times n$ matrix, then $\operatorname{row} A = \operatorname{col} A$ because they both have the same dimension.

☐ True

☐ False

- (e) If $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a linearly independent subset of $\mathbf{P}_3$, then $\{\mathbf{x}, \mathbf{y}, \mathbf{x} + \mathbf{y} + \mathbf{z}\}$ is also linearly independent.

☐ True

☐ False

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